

Utility Crystal Ball

Model Card

Forecasting Engine Documentation

Platform	Utility Crystal Ball
Version	2.0 — Daily + Monthly + Temperature Mode
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Model type	Ensemble forecasting (Prophet + ETS)
Supported frequencies	Monthly Daily
Supported utilities	Electricity, Water, Gas, District Cooling, TSE
Temperature enhancement	Optional — Electricity & District Cooling

1. Platform Overview

Utility Crystal Ball is a professional AI-powered forecasting platform designed for utility planners, asset managers, and operations engineers. It provides multi-scenario consumption forecasts for five utility types using an ensemble of state-of-the-art time series models, with optional temperature enhancement for improved accuracy on climate-sensitive utilities.

Intended use cases:

- Capital planning — justify infrastructure investment with data-driven demand forecasts
- Asset management — anticipate consumption stress before assets reach capacity limits
- Operations planning — schedule maintenance around predicted peak and low periods
- Sustainability reporting — forecast demand reduction opportunities and TSE utilization
- Tariff and procurement — negotiate contracts based on predicted future demand
- Risk management — scenario planning with optimistic, base, and conservative forecasts

2. Model Architecture — Standard Mode (No Temperature)

The standard forecasting engine uses a two-model ensemble combining Prophet and Exponential Smoothing (ETS). Each model contributes different strengths, and the ensemble combines them to produce a more robust and accurate forecast than either model alone.

2.1 Prophet Model (70% weight)

Prophet is an open-source forecasting model developed by Meta (Facebook) Research, specifically designed for business time series with strong seasonal patterns. It decomposes the time series into trend, seasonality, and noise components.

- Yearly seasonality — captures annual consumption cycles (summer peaks, winter peaks)
- Weekly seasonality — for daily data, captures Mon-Sun consumption patterns
- Trend detection — identifies long-term growth or decline in consumption
- Changepoint detection — automatically finds moments where trend direction changed
- Multiplicative seasonality — handles data where seasonal amplitude grows with the trend
- Holiday effects — can incorporate calendar events that affect consumption

2.2 ETS Model — Exponential Smoothing (30% weight)

Exponential Smoothing (ETS) is a classical statistical forecasting method that assigns exponentially decreasing weights to past observations. It is particularly effective for stable, smooth time series and provides a natural complement to Prophet.

- Additive trend — models steady linear growth or decline in consumption
- Additive seasonality — captures regular seasonal fluctuations
- Automatic optimization — parameters fitted to minimize forecast error on training data
- Stability — produces smooth forecasts without wild oscillations or spikes
- Short-term accuracy — typically outperforms Prophet on near-term predictions

2.3 Ensemble Combination

The final forecast is a weighted average of both models: Prophet contributes 70% and ETS contributes 30%. This weighting reflects the superior seasonal pattern detection of Prophet for utility data, while preserving the smoothing benefits of ETS.

Component	Weight	Primary Strength
Prophet	70%	Seasonality, trend, changepoints
ETS	30%	Smoothness, stability, short-term
Ensemble output	100%	Combined forecast (base case)

3. Temperature-Enhanced Mode (New)

For electricity and district cooling forecasting, temperature is the single most important external driver of consumption. The temperature-enhanced mode extends the standard ensemble by incorporating daily maximum and minimum temperature as explicit regressors in the Prophet model.

3.1 Philosophy

Standard mode learns that "July is always high" from historical patterns alone. Temperature mode learns the physical causal relationship: "when temperature reaches 44 degrees Celsius, consumption increases by X MWh due to air conditioning load." This is a fundamentally more accurate representation of the underlying physical reality.

3.2 How it works

- User uploads historical data with Temp_Max_C and Temp_Min_C columns alongside consumption
- Prophet treats temperature as an additive regressor — learning the consumption-temperature coefficient
- For the forecast period, user provides future temperature forecast (from Open-Meteo or similar)
- Days 1-16: real meteorological forecast (Open-Meteo free API — 16 days ahead)
- Days 17-30: seasonal historical average from the same month in prior year(s)

- Prophet applies the learned temperature coefficient to project future consumption

3.3 Temperature data source

Historical temperature: Open-Meteo Historical Weather API (free, no registration required). Future temperature forecast: Open-Meteo Forecast API (free tier provides 16 days ahead). Beyond 16 days: platform automatically calculates seasonal average from historical data. Coordinates used for UAE: Latitude 24.4539, Longitude 54.3773 (Abu Dhabi).

3.4 Expected accuracy improvement

Mode	Expected MAPE	Conditions
Standard (no temperature)	75-90%	Monthly or daily, sufficient history
Temperature-enhanced	85-95%	Daily data, temperature provided
Temperature — short term (7 days)	90-97%	Near-term daily with real forecast

4. Model Validation — Walk-Forward Backtest

Every forecast run includes an automatic walk-forward backtest to measure real predictive accuracy. The last 10% of historical data is withheld from training and used exclusively for validation.

4.1 Backtest methodology

- Training set: first 90% of historical data
- Test set: last 10% (minimum 12 months for monthly, minimum 30 days for daily)
- Model trained on training set only — test set never seen during training
- Predictions made for entire test period
- Predicted values compared against actual values
- MAPE calculated as: 100% minus mean absolute percentage error

4.2 Accuracy metrics

Metric	Definition	Interpretation
MAPE	Mean Absolute % Error accuracy	Higher = better. 90%+ = excellent
MAE	Mean Absolute Error in original units	Lower = better. Context-dependent
Trend %	Forecast end vs last historical value	Direction and magnitude of change

5. Data Requirements

Parameter	Monthly Mode	Daily Mode
Minimum history	60 months	365 days
Recommended history	120+ months	730+ days
Date format	YYYY-MM-DD	YYYY-MM-DD
Frequency	First day of each month	One row per day

Missing values	Below 5% recommended	Below 5% recommended
Forecast horizons	12 / 24 / 36 / 60 months	30 / 60 / 90 / 180 days
Temperature columns	Optional	Temp_Max_C + Temp_Min_C
Multi-zone support	Unlimited columns	Single or multiple columns

6. Scenario Forecasting

The platform generates three forecast scenarios simultaneously, providing planners with a range of possible futures for robust decision-making rather than a single point estimate.

Scenario	Method	Use case
Base case	Ensemble forecast (Prophet 70% + ETS 30%)	Most likely outcome — primary planning
Optimistic	Base case + 6%	Lower demand — best case infrastructure load
Conservative	Base case - 6%	Higher demand — worst case capacity planning

7. Limitations and Honest Caveats

- Daily forecasts beyond 30 days have significantly reduced accuracy — use monthly mode for longer horizons
- Models cannot predict unknown future events (new infrastructure, policy changes, extreme weather anomalies)
- COVID-19 period (2020-2021) distorted consumption patterns — consider excluding from training data
- Temperature mode requires future temperature forecast — accuracy of temperature forecast affects output
- Minimum history requirements are enforced — results with borderline data should be treated with caution
- Models assume future consumption patterns broadly resemble historical patterns
- Unusual spikes in historical data are dampened — model assumes they are anomalies not to be repeated

8. Platform Features Summary

Feature	Description
Data Explorer	Time series, distribution, seasonality, YoY, heatmap, statistics table
Validation gates	3-gate system: structure, quality, UAE-specific checks
Data quality score	Automatic 0-100 score with color coding
Backtest chart	Visual actual vs predicted with error table per period
Seasonality chart	Monthly index + day-of-week pattern (daily mode)
Zone insights	Automatic status, trend, peak month per zone
Excel export	Summary + forecast + backtest + seasonality + YoY sheets
PDF report	Professional branded report with all tables and methodology
AI insights	Optional Anthropic API key for executive narrative generation
Sample files	Clean and intentionally flawed samples for testing
About page	Platform documentation accessible from landing page

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This document — accessible from platform

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